INCH-POUND
MIL-DTL-24231/8E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/8D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTORS, SINGLE CABLE, TYPE III, SEVEN-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 5.

2. Refer to MIL-DTL-24231/5 for alternate method of manufacturing connector body and installing guide pin M24231/5-071.

3. For welding of the pin, item 20, see MIL-DTL-24231/5.

4. Full or partial threads shall not extend more than 0.083 inch beyond the 11/16-inch thread length.

5. All dimensions are in inches.

6. Tolerances shall be as follows, or as stated:

   a. Decimals – ±0.005 inch.

   b. Fractions – ±1/64 inch.

   c. Angular dimensions – ±½ degree.

7. The flange outside diameter is intended to coincide with commonly available 2¾-inch bar stock. The dimension may vary after machine clean-up.

8. The gland nut shall be tightened until the packing tightens against the wires and cables. This can be determined by moving the wires and cables gently back and forth until the wire and cable is seized by the packing. The gland nut shall be turned approximately one-fourth of a turn more to hold wires and cables in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires and cables.

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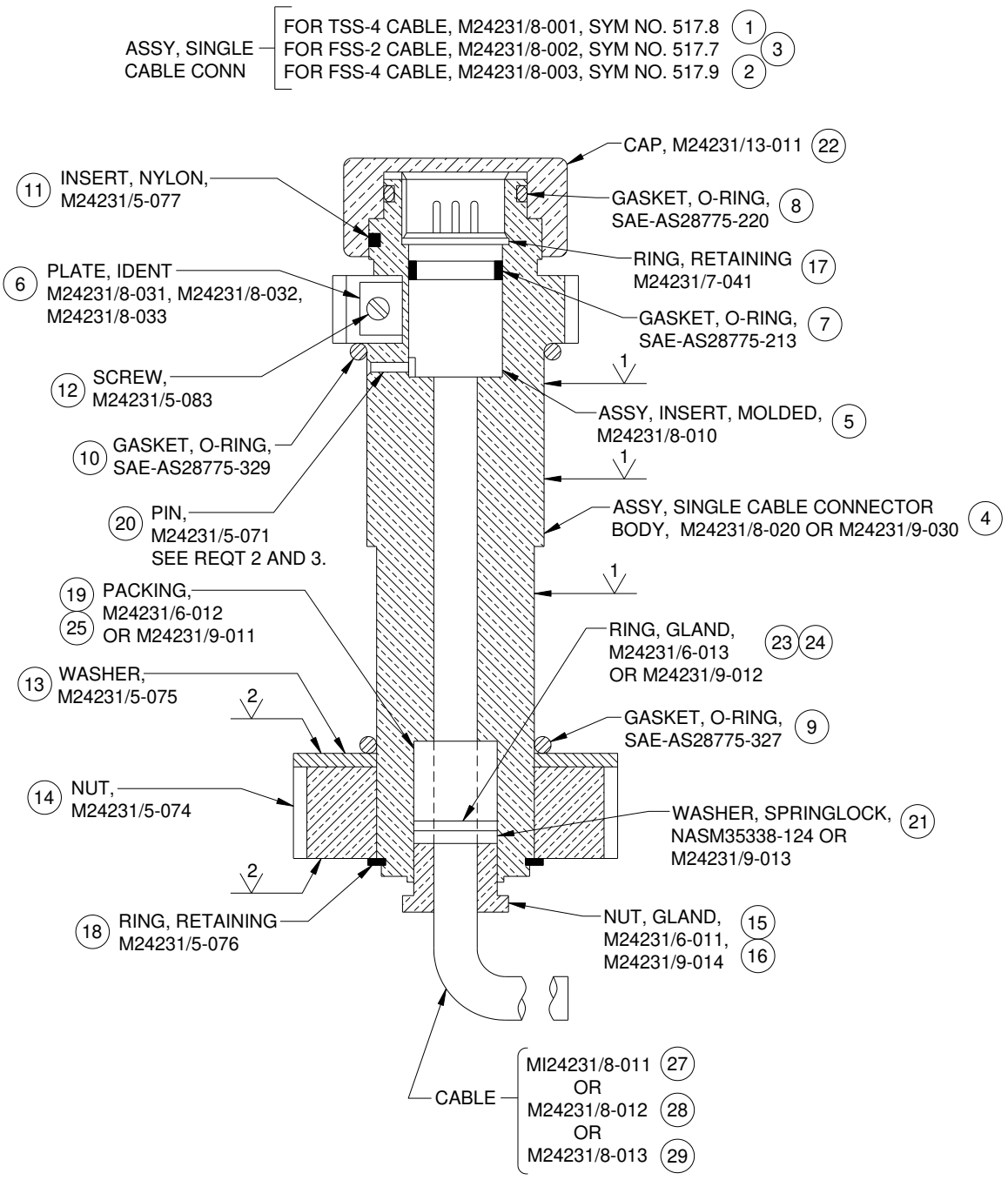


FIGURE 1. Connector, single cable, seven-conductor.

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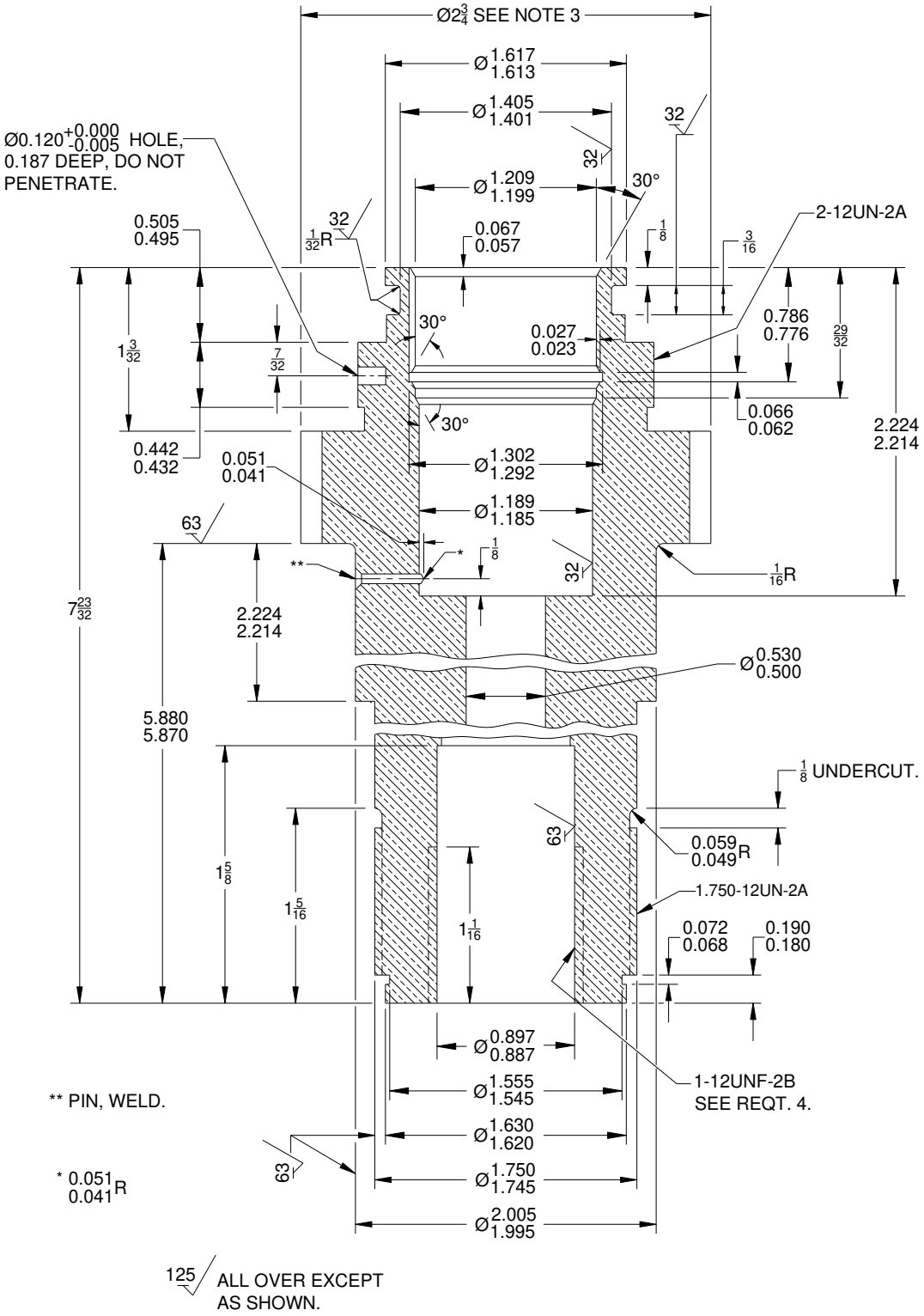


FIGURE 2. Assembly, single cable connector body – M24231/8-020 (for M24231/8-001 and M24231/8-002 only).

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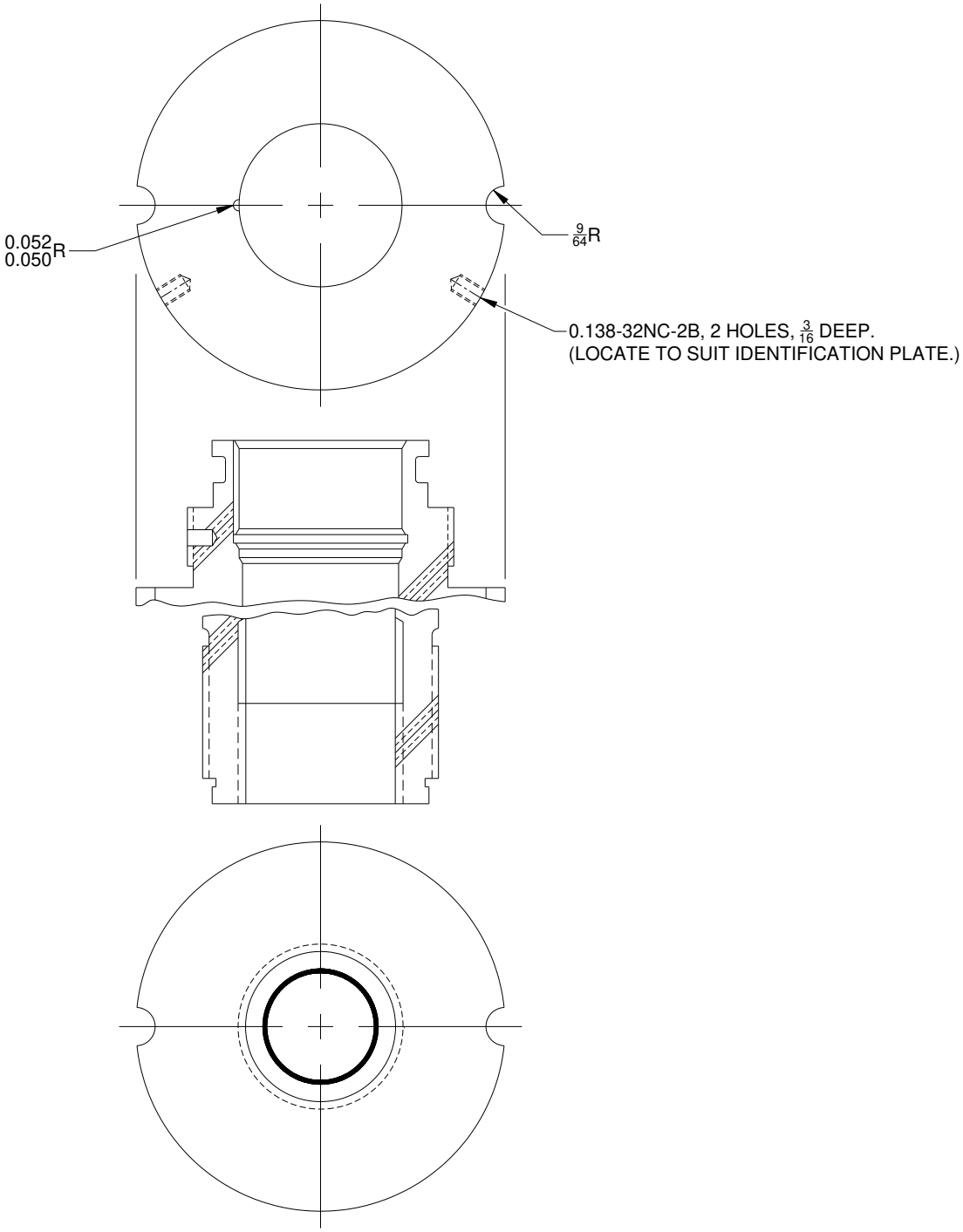


FIGURE 3. Assembly, single cable connector body – M24231/8-020.

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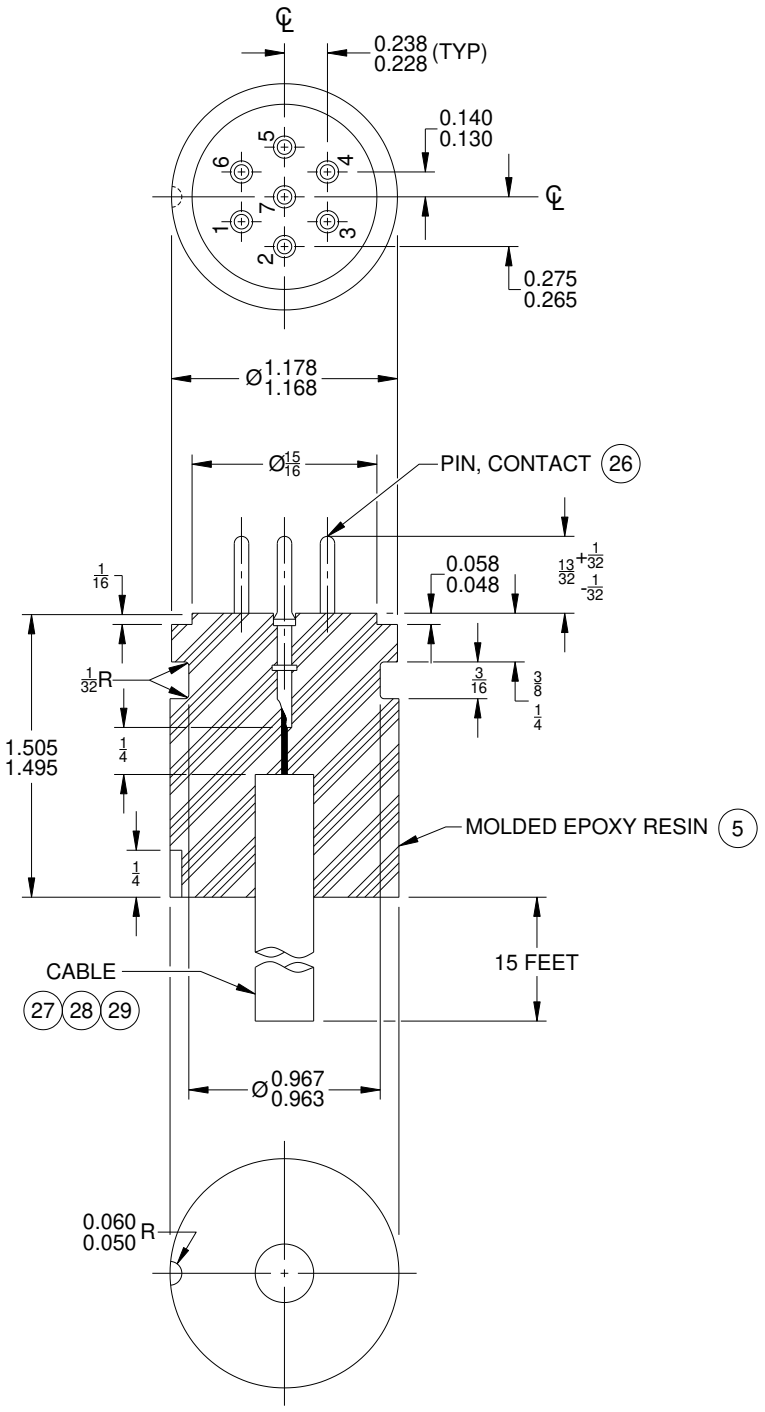


FIGURE 4. Assembly, insert, molded – M24231/8-010. 30

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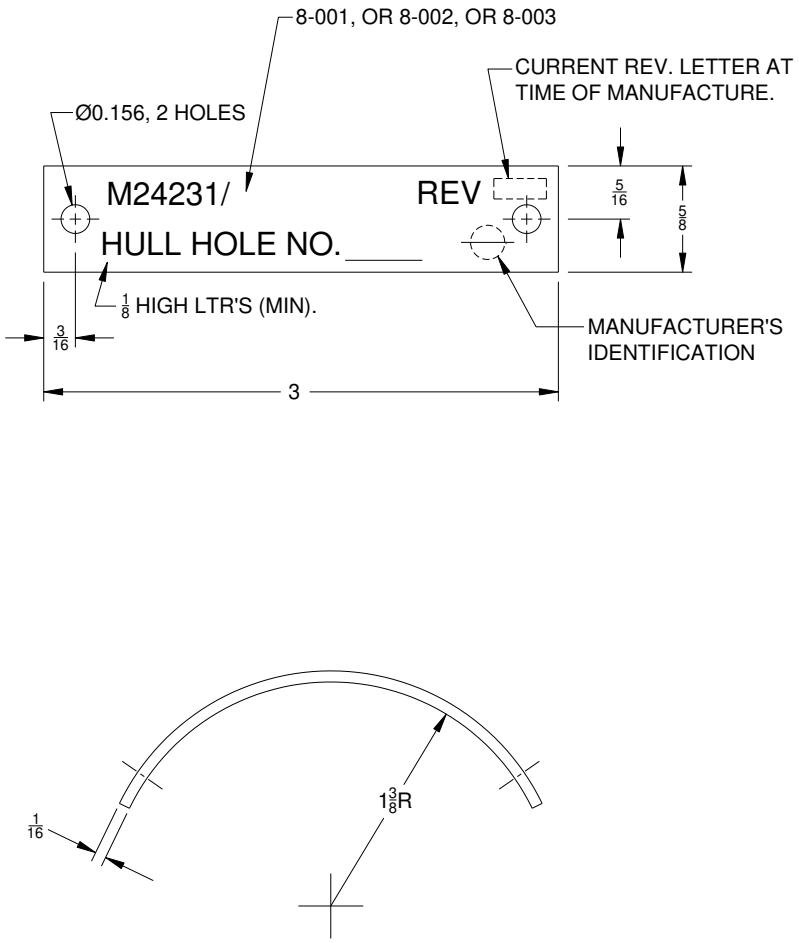


FIGURE 5. Plate, identification – M24231/8-031 (for M24231/8-001), M24231/8-032 (for M24231/8-002), M24231/8-033 (for M24231/8-003). (6)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/8- 3/			Material/specification	Remarks
		001	002	003		
1	Connector assembly, single cable, symbol no. 517.8	1			-----	See figure 1.
2	Connector assembly, single cable, symbol no. 517.7		1		-----	See figure 1.
3	Connector assembly, single cable, symbol 517.9			1	-----	See figure 1.
4	Connector, body, single cable	1	1		Ni-Cu-Al, QQ-N-286	See figure 2 and figure 3 and requirement 2.
5	Epoxy resin, molded	As required			Cast epoxy resin compound 4/	See figure 4.
6	Plate, identification	1	1	1	CRES or Al, MIL-DTL-15024	See figure 5.
7	Gasket, O-ring	1	1	1	SAE-AS28775-213 5/	Lubricate with silicone compound SAE-AS8660, see figure 1.
8	Gasket, O-ring	1	1	1	SAE-AS28775-220 5/	See item 7.
9	Gasket, O-ring	1	1	1	SAE-AS28775-327 5/	See item 7.
10	Gasket, O-ring	1	1	1	SAE-AS28775-329 5/	See item 7.
11	Insert	1	1	1	M24231/5-077	See figure 1.
12	Screw	2	2	2	M24231/5-083	See figure 1.
13	Washer, body	1	1	1	M24231/5-075	See figure 1.
14	Nut, body	1	1	1	M24231/5-074	See figure 1.
15	Nut, gland	1	1		M24231/6-011	See figure 1.
16	Nut, gland			1	M24231/9-014	See figure 1.
17	Ring, retaining	1	1	1	M24231/7-041	See figure 1.
18	Ring, retaining	1	1	1	M24231/5-076	See figure 1.
19	Packing	1	1		M24231/6-012	See figure 1.
20	Pin, guide	1	1	1	M24231/5-071	See requirement 2.
21	Washer, spring-lock	1	1	1	NASM35338-124	See figure 1.
22	Cap, pressure-proof	1	1	1	M24231/13-011	See figure 1.
23	Ring, gland	1	1		M24231/6-013	See figure 1.
24	Ring, gland			1	M24231/9-012	See figure 1.
25	Packing			1	M24231/9-011	See figure 1.
26	Pin, contact	7	7	7	MIL-DTL-24231	See figure 4.
27	Cable, TSS-4, 15’	1			MIL-DTL-915/8	See figure 1 and figure 4.
28	Cable, FSS-2, 15’		1		MIL-DTL-915/8	See figure 1 and figure 4.
29	Cable, FSS-4, 15’			1	MIL-DTL-915/8	See figure 1 and figure 4.

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TABLE I. Parts list – Continued.

Item no. 1/	Description 2/	Quantity for M24231/8- 3/			Material/Specification	Remarks
		001	002	003		
30	Insert assembly, molded	1	1	1	See items 05, 26 and 27, 28, or 29	See figure 4.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Arco Set 200, M.M. Reif and Co., Ivyland, PA, or PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

5/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-057)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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